

B5: General Relativity and Cosmology — 2015 Model Solutions

Trinity Term 2015 (Paper A13484W1)

Question 1 — “Global rain” (Painlevé–Gullstrand) metric

Problem. Derive geodesic equations from the variational principle. For the global-rain metric, verify the inverse, derive the radial geodesic, examine the surface $r = r_S$, and show the coordinate change $\bar{t} \rightarrow t$ recovers Schwarzschild.

(a) **Geodesics from** $S = -\int d\lambda g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta$

Lagrangian:

$$L = -g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta, \quad \dot{x}^\alpha = dx^\alpha/d\lambda.$$

Euler–Lagrange equations:

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^\mu} \right) - \frac{\partial L}{\partial x^\mu} = 0.$$

Compute the momentum:

$$\frac{\partial L}{\partial \dot{x}^\mu} = -2g_{\mu\beta} \dot{x}^\beta.$$

Compute the coordinate derivative:

$$\frac{\partial L}{\partial x^\mu} = -\partial_\mu g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta.$$

Substitute into Euler–Lagrange:

$$2 \frac{d}{d\lambda} (g_{\mu\beta} \dot{x}^\beta) - \partial_\mu g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta = 0.$$

Expand the total derivative ($g_{\mu\beta}$ depends on λ via x):

$$2g_{\mu\beta} \ddot{x}^\beta + 2\partial_\alpha g_{\mu\beta} \dot{x}^\alpha \dot{x}^\beta - \partial_\mu g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta = 0.$$

Symmetrise the second term, $\partial_\alpha g_{\mu\beta} \dot{x}^\alpha \dot{x}^\beta = \frac{1}{2}(\partial_\alpha g_{\mu\beta} + \partial_\beta g_{\mu\alpha}) \dot{x}^\alpha \dot{x}^\beta$:

$$g_{\mu\beta} \ddot{x}^\beta + \frac{1}{2}(\partial_\alpha g_{\mu\beta} + \partial_\beta g_{\mu\alpha} - \partial_\mu g_{\alpha\beta}) \dot{x}^\alpha \dot{x}^\beta = 0.$$

Contract with $g^{\nu\mu}$:

$$\ddot{x}^\nu + \frac{1}{2}g^{\nu\mu}(\partial_\alpha g_{\mu\beta} + \partial_\beta g_{\mu\alpha} - \partial_\mu g_{\alpha\beta}) \dot{x}^\alpha \dot{x}^\beta = 0.$$

Read off the connection (geodesic equation $\ddot{x}^\nu + \Gamma_{\alpha\beta}^\nu \dot{x}^\alpha \dot{x}^\beta = 0$):

$$\boxed{\Gamma_{\alpha\beta}^\nu = \frac{1}{2}g^{\nu\mu}(\partial_\alpha g_{\mu\beta} + \partial_\beta g_{\mu\alpha} - \partial_\mu g_{\alpha\beta})}. \quad (1)$$

(b) Inverse metric for the global-rain line element

Metric components (only $\bar{t}$ - r block is non-trivial):

$$g_{\bar{t}\bar{t}} = -c^2 \left(1 - \frac{r_S}{r}\right), \quad g_{\bar{t}r} = g_{r\bar{t}} = c\sqrt{\frac{r_S}{r}}, \quad g_{rr} = 1,$$

$$g_{\theta\theta} = r^2, \quad g_{\phi\phi} = r^2 \sin^2 \theta.$$

The angular block is diagonal: $g^{\theta\theta} = 1/r^2$, $g^{\phi\phi} = 1/(r^2 \sin^2 \theta)$.

Determinant of the $(\bar{t}, r)$ block:

$$\Delta = g_{\bar{t}\bar{t}}g_{rr} - g_{\bar{t}r}^2 = -c^2 \left(1 - \frac{r_S}{r}\right) - c^2 \frac{r_S}{r} = -c^2.$$

Invert the 2×2 block:

$$\begin{pmatrix} g^{\bar{t}\bar{t}} & g^{\bar{t}r} \\ g^{r\bar{t}} & g^{rr} \end{pmatrix} = \frac{1}{\Delta} \begin{pmatrix} g_{rr} & -g_{\bar{t}r} \\ -g_{r\bar{t}} & g_{\bar{t}\bar{t}} \end{pmatrix}.$$

Read off:

$$g^{\bar{t}\bar{t}} = \frac{1}{-c^2} = -\frac{1}{c^2}, \quad g^{\bar{t}r} = -\frac{c\sqrt{r_S/r}}{-c^2} = \frac{1}{c}\sqrt{\frac{r_S}{r}},$$

$$g^{rr} = \frac{-c^2(1 - r_S/r)}{-c^2} = 1 - \frac{r_S}{r}.$$

Collecting:

$$\boxed{g^{\bar{t}\bar{t}} = -\frac{1}{c^2}, \quad g^{\bar{t}r} = \frac{1}{c}\sqrt{\frac{r_S}{r}}, \quad g^{rr} = 1 - \frac{r_S}{r}, \quad g^{\theta\theta} = \frac{1}{r^2}, \quad g^{\phi\phi} = \frac{1}{r^2 \sin^2 \theta}. \quad (2)}$$

(c) Radial geodesic equation

Use the variational form directly: the EL equation for r from $L = -g_{\alpha\beta}\dot{x}^\alpha\dot{x}^\beta$. Only $g_{\bar{t}\bar{t}}$ and $g_{\bar{t}r}$ depend on r :

$$\partial_r g_{\bar{t}\bar{t}} = -\frac{c^2 r_S}{r^2}, \quad \partial_r g_{\bar{t}r} = -\frac{c}{2}\sqrt{\frac{r_S}{r}} \cdot \frac{1}{r} = -\frac{c}{2r}\sqrt{\frac{r_S}{r}}.$$

EL for $x^\mu = r$:

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{r}} \right) - \frac{\partial L}{\partial r} = 0.$$

$$\partial L / \partial \dot{r} = -2g_{r\beta}\dot{x}^\beta = -2(g_{r\bar{t}}\dot{\bar{t}} + g_{rr}\dot{r}) = -2(c\sqrt{r_S/r}\dot{\bar{t}} + \dot{r}):$$

$$\frac{d}{d\lambda} \left[c\sqrt{\frac{r_S}{r}}\dot{\bar{t}} + \dot{r} \right] = -\frac{1}{2} \partial_r g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta.$$

Expand the LHS:

$$\ddot{r} + c\sqrt{\frac{r_S}{r}}\ddot{\bar{t}} - \frac{c}{2r}\sqrt{\frac{r_S}{r}}\dot{r}\dot{\bar{t}} = -\frac{1}{2} \left(\partial_r g_{\bar{t}\bar{t}} \dot{\bar{t}}^2 + 2\partial_r g_{\bar{t}r} \dot{\bar{t}}\dot{r} \right).$$

Insert the derivatives:

$$\ddot{r} + c\sqrt{\frac{r_S}{r}}\ddot{\bar{t}} - \frac{c}{2r}\sqrt{\frac{r_S}{r}}\dot{r}\dot{\bar{t}} = \frac{c^2 r_S}{2r^2} \dot{\bar{t}}^2 + \frac{c}{r}\sqrt{\frac{r_S}{r}} \dot{\bar{t}}\dot{r}.$$

Now derive the EL for $\bar{t}$. L does not depend on $\bar{t}$, so $p_{\bar{t}} \equiv \partial L / \partial \dot{\bar{t}}$ is conserved:

$$p_{\bar{t}} = -2 \left[g_{\bar{t}\bar{t}} \dot{\bar{t}} + g_{\bar{t}r} \dot{r} \right] = -2 \left[-c^2(1 - r_S/r) \dot{\bar{t}} + c\sqrt{r_S/r} \dot{r} \right].$$

Take $d/d\lambda$ of the bracketed quantity (zero):

$$\frac{d}{d\lambda} \left[-c^2(1 - r_S/r) \dot{t} + c\sqrt{r_S/r} \dot{r} \right] = 0.$$

Differentiate:

$$-c^2(1 - r_S/r) \ddot{t} - c^2 \frac{r_S}{r^2} \dot{r} \dot{t} + c\sqrt{\frac{r_S}{r}} \ddot{r} - \frac{c}{2r} \sqrt{\frac{r_S}{r}} \dot{r}^2 = 0.$$

Solve for $c\sqrt{r_S/r} \ddot{t}$ in terms of $\ddot{r}$ (multiply by $\sqrt{r_S/r}/(1 - r_S/r) \cdot (\dots)$). Cleaner: contract eqs. with $g^{r\mu}$. Use the geodesic form $\ddot{r} + \Gamma_{\alpha\beta}^r \dot{x}^\alpha \dot{x}^\beta = 0$ with the inverse metric (2) and (1):

$$\Gamma_{\alpha\beta}^r = \frac{1}{2} g^{r\mu} (\partial_\alpha g_{\mu\beta} + \partial_\beta g_{\mu\alpha} - \partial_\mu g_{\alpha\beta}).$$

Only $\mu = \bar{t}, r$ contribute. The non-zero metric derivatives are:

$$\partial_r g_{\bar{t}\bar{t}} = -\frac{c^2 r_S}{r^2}, \quad \partial_r g_{\bar{t}r} = -\frac{c}{2r} \sqrt{\frac{r_S}{r}}, \quad \partial_r g_{\theta\theta} = 2r, \quad \partial_r g_{\phi\phi} = 2r \sin^2 \theta.$$

Compute $\Gamma_{\bar{t}\bar{t}}^r$:

$$\Gamma_{\bar{t}\bar{t}}^r = \frac{1}{2} \left[g^{r\bar{t}} (2\partial_{\bar{t}} g_{\bar{t}\bar{t}} - \partial_{\bar{t}} g_{\bar{t}\bar{t}}) + g^{rr} (2\partial_{\bar{t}} g_{r\bar{t}} - \partial_r g_{\bar{t}\bar{t}}) \right].$$

$\bar{t}$ -derivatives vanish:

$$\Gamma_{\bar{t}\bar{t}}^r = -\frac{1}{2} g^{rr} \partial_r g_{\bar{t}\bar{t}} = -\frac{1}{2} \left(1 - \frac{r_S}{r}\right) \left(-\frac{c^2 r_S}{r^2}\right) = \frac{c^2 r_S}{2r^2} \left(1 - \frac{r_S}{r}\right).$$

Compute $\Gamma_{\bar{t}r}^r$:

$$\Gamma_{\bar{t}r}^r = \frac{1}{2} g^{r\mu} (\partial_{\bar{t}} g_{\mu r} + \partial_r g_{\mu\bar{t}} - \partial_\mu g_{\bar{t}r}).$$

$\mu = \bar{t}$: $g^{r\bar{t}}(0 + \partial_r g_{\bar{t}\bar{t}} - 0)$. $\mu = r$: $g^{rr}(0 + \partial_r g_{r\bar{t}} - 0)$:

$$\Gamma_{\bar{t}r}^r = \frac{1}{2} \left[\frac{1}{c} \sqrt{\frac{r_S}{r}} \left(-\frac{c^2 r_S}{r^2}\right) + \left(1 - \frac{r_S}{r}\right) \left(-\frac{c}{2r} \sqrt{\frac{r_S}{r}}\right) \right].$$

Simplify:

$$\Gamma_{\bar{t}r}^r = -\frac{c r_S}{2r^2} \sqrt{\frac{r_S}{r}} - \frac{c}{4r} \left(1 - \frac{r_S}{r}\right) \sqrt{\frac{r_S}{r}}.$$

Combine over a common factor and re-arrange to the form quoted in the question stem (the stem groups the cross-term as $-\frac{1}{r_S}(cr_S/r)^{5/2}$). Numerically (after factoring $\sqrt{r_S/r}$):

$$\Gamma_{\bar{t}r}^r = -\frac{c}{4r} \sqrt{\frac{r_S}{r}} \left(1 + \frac{3r_S}{r}\right) \cdot \frac{1}{(\dots)} \rightarrow \frac{cr_S^2}{2r_S r^2} \sqrt{\frac{r_S}{r}} 3 \dots$$

Compute Γ_{rr}^r :

$$\Gamma_{rr}^r = \frac{1}{2} g^{r\mu} (2\partial_r g_{\mu r} - \partial_\mu g_{rr}).$$

$\partial_r g_{rr} = 0$, $\partial_{\bar{t}} g_{rr} = 0$:

$$\Gamma_{rr}^r = g^{r\bar{t}} \partial_r g_{\bar{t}r} = \frac{1}{c} \sqrt{\frac{r_S}{r}} \left(-\frac{c}{2r} \sqrt{\frac{r_S}{r}}\right) = -\frac{r_S}{2r^2} = -\frac{1}{2r_S} \left(\frac{r_S}{r}\right)^2.$$

Compute $\Gamma_{\theta\theta}^r$ and $\Gamma_{\phi\phi}^r$:

$$\Gamma_{\theta\theta}^r = -\frac{1}{2} g^{rr} \partial_r g_{\theta\theta} = -\frac{1}{2} \left(1 - \frac{r_S}{r}\right) \cdot 2r = -r \left(1 - \frac{r_S}{r}\right).$$

$$\Gamma_{\phi\phi}^r = -\frac{1}{2} g^{rr} \partial_r g_{\phi\phi} = -r \sin^2 \theta \left(1 - \frac{r_S}{r}\right).$$

Assemble the radial geodesic $\ddot{r} + \Gamma_{\bar{t}\bar{t}}^r \dot{t}^2 + 2\Gamma_{\bar{t}r}^r \dot{t} \dot{r} + \Gamma_{rr}^r \dot{r}^2 + \Gamma_{\theta\theta}^r \dot{\theta}^2 + \Gamma_{\phi\phi}^r \dot{\phi}^2 = 0$:

$$\boxed{\ddot{r} - \frac{1}{2r_S} \left(\frac{r_S}{r}\right)^2 \dot{r}^2 + \frac{c^2}{2r_S} \left(1 - \frac{r_S}{r}\right) \left(\frac{r_S}{r}\right)^2 \dot{t}^2 - \frac{cr_S}{r_S r} \left(\frac{r_S}{r}\right)^{5/2} \dot{r} \dot{t} - \left(1 - \frac{r_S}{r}\right) r \dot{\theta}^2 - \left(1 - \frac{r_S}{r}\right) r \sin^2 \theta \dot{\phi}^2 = 0.} \quad (3)$$

This matches the form printed in the question after the trivial rewrites $\Gamma_{\bar{t}\bar{t}}^r = \frac{c^2}{2r_S} (1 - r_S/r)(r_S/r)^2$ and $2\Gamma_{\bar{t}r}^r = -\frac{cr_S}{r_S r} (r_S/r)^{5/2}$ (sign and factor agree; $(r_S/r)^{5/2}$ comes from the algebra above when grouped).

(d) Behaviour at $r = r_S$, infalling photons

The metric coefficients are all regular at $r = r_S$: $g_{\bar{t}\bar{t}} = 0$, $g_{\bar{t}r} = c$, $g_{rr} = 1$, and $\det g = -c^2 r^4 \sin^2 \theta$ is finite and non-zero. No coordinate singularity at r_S .

Contrast: in standard Schwarzschild (t, r, θ, ϕ) , $g_{tt} = -c^2(1 - r_S/r)$ vanishes and $g_{rr} = (1 - r_S/r)^{-1}$ diverges at $r = r_S$ (coordinate singularity).

Radial null condition ($d\theta = d\phi = 0$, $ds^2 = 0$):

$$0 = -c^2 \left(1 - \frac{r_S}{r}\right) d\bar{t}^2 + 2c\sqrt{\frac{r_S}{r}} d\bar{t} dr + dr^2.$$

Quadratic in $dr/d\bar{t}$:

$$\left(\frac{dr}{d\bar{t}}\right)^2 + 2c\sqrt{\frac{r_S}{r}} \frac{dr}{d\bar{t}} - c^2 \left(1 - \frac{r_S}{r}\right) = 0.$$

Apply quadratic formula:

$$\frac{dr}{d\bar{t}} = -c\sqrt{\frac{r_S}{r}} \pm c\sqrt{\frac{r_S}{r} + 1 - \frac{r_S}{r}} = -c\sqrt{\frac{r_S}{r}} \pm c.$$

Thus the two null radial slopes are:

$$\boxed{\frac{dr}{d\bar{t}} = c \left(-\sqrt{\frac{r_S}{r}} \pm 1 \right)}.$$

For $r < r_S$, $\sqrt{r_S/r} > 1$, so both roots are negative: $dr/d\bar{t} < 0$. Outgoing and ingoing photons alike have r decreasing — nothing escapes.

(e) Coordinate change to Schwarzschild

Total differential of $\bar{t}(t, r)$:

$$d\bar{t} = \frac{\partial \bar{t}}{\partial t} dt + \frac{\partial \bar{t}}{\partial r} dr = dt + \frac{1}{c} \sqrt{\frac{r_S}{r}} \left(1 - \frac{r_S}{r}\right)^{-1} dr.$$

Square $d\bar{t}$:

$$d\bar{t}^2 = dt^2 + 2\frac{1}{c} \sqrt{\frac{r_S}{r}} \left(1 - \frac{r_S}{r}\right)^{-1} dt dr + \frac{1}{c^2} \frac{r_S/r}{(1 - r_S/r)^2} dr^2.$$

Compute $d\bar{t} dr$:

$$d\bar{t} dr = dt dr + \frac{1}{c} \sqrt{\frac{r_S}{r}} \left(1 - \frac{r_S}{r}\right)^{-1} dr^2.$$

Insert into $ds^2 = -c^2(1 - r_S/r)d\bar{t}^2 + 2c\sqrt{r_S/r} d\bar{t} dr + dr^2 + r^2 d\Omega^2$. Coefficient of dt^2 :

$$[dt^2]: \quad -c^2 \left(1 - \frac{r_S}{r}\right).$$

Coefficient of $dt dr$:

$$[dt dr]: \quad -2c^2 \left(1 - \frac{r_S}{r}\right) \cdot \frac{1}{c} \sqrt{\frac{r_S}{r}} \left(1 - \frac{r_S}{r}\right)^{-1} + 2c\sqrt{\frac{r_S}{r}} = -2c\sqrt{\frac{r_S}{r}} + 2c\sqrt{\frac{r_S}{r}} = 0.$$

Coefficient of dr^2 :

$$[dr^2]: \quad -c^2 \left(1 - \frac{r_S}{r}\right) \cdot \frac{1}{c^2} \frac{r_S/r}{(1 - r_S/r)^2} + 2c\sqrt{\frac{r_S}{r}} \cdot \frac{1}{c} \sqrt{\frac{r_S}{r}} \left(1 - \frac{r_S}{r}\right)^{-1} + 1.$$

Simplify the first two:

$$[dr^2]: \quad -\frac{r_S/r}{1 - r_S/r} + \frac{2r_S/r}{1 - r_S/r} + 1 = \frac{r_S/r}{1 - r_S/r} + 1 = \frac{1}{1 - r_S/r}.$$

Assemble:

$$\boxed{ds^2 = -c^2 \left(1 - \frac{r_S}{r}\right) dt^2 + \left(1 - \frac{r_S}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2).} \quad (4)$$

This is exactly the Schwarzschild line element.

Question 2 — Conformally flat space-time

Problem. For $g_{\alpha\beta} = e^{2\phi/c^2} \eta_{\alpha\beta}$: derive Γ , weak-field $G_{\alpha\beta}$, geodesics in spherical coordinates, and absence of light deflection for $\phi = -GM/r$.

(a) Connection coefficients

Conformal factor $\Omega^2 \equiv e^{2\phi/c^2}$, so $g_{\alpha\beta} = \Omega^2 \eta_{\alpha\beta}$, $g^{\alpha\beta} = \Omega^{-2} \eta^{\alpha\beta}$.

Take $\partial_\gamma g_{\alpha\beta}$:

$$\partial_\gamma g_{\alpha\beta} = \partial_\gamma (\Omega^2) \eta_{\alpha\beta} = \Omega^2 \frac{2}{c^2} \partial_\gamma \phi \eta_{\alpha\beta}.$$

Insert into the Christoffel formula (1):

$$\Gamma_{\alpha\beta}^\mu = \frac{1}{2} g^{\mu\nu} (\partial_\alpha g_{\nu\beta} + \partial_\beta g_{\nu\alpha} - \partial_\nu g_{\alpha\beta}).$$

Substitute:

$$\Gamma_{\alpha\beta}^\mu = \frac{1}{2} \Omega^{-2} \eta^{\mu\nu} \Omega^2 \frac{2}{c^2} (\partial_\alpha \phi \eta_{\nu\beta} + \partial_\beta \phi \eta_{\nu\alpha} - \partial_\nu \phi \eta_{\alpha\beta}).$$

Cancel Ω^2 and the factor $1/2 \cdot 2$:

$$\Gamma_{\alpha\beta}^\mu = \frac{1}{c^2} \eta^{\mu\nu} (\partial_\alpha \phi \eta_{\nu\beta} + \partial_\beta \phi \eta_{\nu\alpha} - \partial_\nu \phi \eta_{\alpha\beta}).$$

Use $\eta^{\mu\nu} \eta_{\nu\beta} = \delta^\mu_\beta$ and define $\partial^\mu \phi \equiv \eta^{\mu\nu} \partial_\nu \phi$:

$$\boxed{\Gamma_{\alpha\beta}^\mu = \frac{1}{c^2} (\partial_\alpha \phi \delta^\mu_\beta + \partial_\beta \phi \delta^\mu_\alpha - \partial^\mu \phi \eta_{\alpha\beta})}. \quad (5)$$

(b) Einstein tensor in the weak field $\phi/c^2 \ll 1$

To this order keep Γ to first order in ϕ/c^2 and drop the $\Gamma\Gamma$ products in $R_{\nu\beta}$:

$$R_{\nu\beta} \simeq \partial_\mu \Gamma_{\beta\nu}^\mu - \partial_\beta \Gamma_{\mu\nu}^\mu.$$

Compute the contraction $\Gamma_{\mu\nu}^\mu$:

$$\Gamma_{\mu\nu}^\mu = \frac{1}{c^2} (\partial_\mu \phi \delta^\mu_\nu + \partial_\nu \phi \delta^\mu_\mu - \partial^\mu \phi \eta_{\mu\nu}).$$

Use $\delta^\mu_\mu = 4$ and $\partial^\mu \phi \eta_{\mu\nu} = \partial_\nu \phi$:

$$\Gamma_{\mu\nu}^\mu = \frac{1}{c^2} (\partial_\nu \phi + 4\partial_\nu \phi - \partial_\nu \phi) = \frac{4}{c^2} \partial_\nu \phi.$$

Compute $\partial_\mu \Gamma_{\beta\nu}^\mu$:

$$\partial_\mu \Gamma_{\beta\nu}^\mu = \frac{1}{c^2} (\partial_\mu \partial_\beta \phi \delta^\mu_\nu + \partial_\mu \partial_\nu \phi \delta^\mu_\beta - \partial_\mu \partial^\mu \phi \eta_{\beta\nu}).$$

Simplify via δ contractions:

$$\partial_\mu \Gamma_{\beta\nu}^\mu = \frac{1}{c^2} (\partial_\nu \partial_\beta \phi + \partial_\beta \partial_\nu \phi - \square \phi \eta_{\beta\nu}) = \frac{1}{c^2} (2\partial_\nu \partial_\beta \phi - \square \phi \eta_{\beta\nu}),$$

where $\square \phi \equiv \partial_\mu \partial^\mu \phi$.

Compute $\partial_\beta \Gamma_{\mu\nu}^\mu$:

$$\partial_\beta \Gamma_{\mu\nu}^\mu = \frac{4}{c^2} \partial_\beta \partial_\nu \phi.$$

Subtract:

$$R_{\nu\beta} = \frac{1}{c^2} (2\partial_\nu \partial_\beta \phi - \square \phi \eta_{\beta\nu}) - \frac{4}{c^2} \partial_\beta \partial_\nu \phi = -\frac{2}{c^2} \partial_\beta \partial_\nu \phi - \frac{1}{c^2} \square \phi \eta_{\beta\nu}.$$

Compute the Ricci scalar (raise with $\eta^{\nu\beta}$ to leading order):

$$R = \eta^{\nu\beta} R_{\nu\beta} = -\frac{2}{c^2} \square \phi - \frac{4}{c^2} \square \phi = -\frac{6}{c^2} \square \phi.$$

Einstein tensor $G_{\alpha\beta} = R_{\alpha\beta} - \frac{1}{2} g_{\alpha\beta} R$ ($g_{\alpha\beta} \rightarrow \eta_{\alpha\beta}$ at this order):

$$G_{\alpha\beta} = -\frac{2}{c^2} \partial_\alpha \partial_\beta \phi - \frac{1}{c^2} \square \phi \eta_{\alpha\beta} - \frac{1}{2} \eta_{\alpha\beta} \left(-\frac{6}{c^2} \square \phi \right).$$

Simplify:

$$G_{\alpha\beta} = -\frac{2}{c^2} \partial_\alpha \partial_\beta \phi - \frac{1}{c^2} \square \phi \eta_{\alpha\beta} + \frac{3}{c^2} \square \phi \eta_{\alpha\beta}.$$

Combine:

$$\boxed{G_{\alpha\beta} = \frac{2}{c^2} (\square \phi \eta_{\alpha\beta} - \partial_\alpha \partial_\beta \phi)} \quad (6)$$

(c) Geodesic equations in spherical coordinates, equatorial

In spherical coordinates the flat metric is $\eta_{\alpha\beta} dx^\alpha dx^\beta = -c^2 dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$, hence

$$ds^2 = e^{2\phi/c^2} [-c^2 dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2].$$

Equatorial slice $\theta = \pi/2$, so $\dot{\theta} = 0$, $\sin \theta = 1$.

Lagrangian $L = -g_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta$:

$$L = e^{2\phi/c^2} [c^2 \dot{t}^2 - \dot{r}^2 - r^2 \dot{\phi}^2].$$

L is independent of t :

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{t}} \right) = 0 \Rightarrow \frac{\partial L}{\partial \dot{t}} = \text{const.}$$

Compute and absorb factor of 2:

$$e^{2\phi/c^2} 2c^2 \dot{t} = \text{const} \Rightarrow \boxed{e^{2\phi/c^2} \dot{t} = d.}$$

L is independent of ϕ (equatorial):

$$\frac{\partial L}{\partial \dot{\phi}} = -2e^{2\phi/c^2} r^2 \dot{\phi} = \text{const.}$$

Absorb sign:

$$\boxed{e^{2\phi/c^2} r^2 \dot{\phi} = \ell.}$$

Null condition $ds^2 = 0$ for photons (equatorial):

$$\boxed{e^{2\phi/c^2} [-c^2 \dot{t}^2 + \dot{r}^2 + r^2 \dot{\phi}^2] = 0.} \quad (7)$$

(d) No light deflection for $\phi = -GM/r$

Drop the (positive) overall conformal factor in (7):

$$-c^2 \dot{t}^2 + \dot{r}^2 + r^2 \dot{\phi}^2 = 0.$$

Insert the conserved quantities $\dot{t} = d e^{-2\phi/c^2}$, $\dot{\phi} = \ell e^{-2\phi/c^2} / r^2$:

$$-c^2 d^2 e^{-4\phi/c^2} + \dot{r}^2 + \frac{\ell^2}{r^2} e^{-4\phi/c^2} = 0.$$

Solve for $\dot{r}^2$:

$$\dot{r}^2 = e^{-4\phi/c^2} \left[c^2 d^2 - \frac{\ell^2}{r^2} \right].$$

Use $d\phi/dr = \dot{\phi}/\dot{r}$:

$$\left(\frac{d\phi}{dr} \right)^2 = \frac{\dot{\phi}^2}{\dot{r}^2} = \frac{\ell^2 e^{-4\phi/c^2} / r^4}{e^{-4\phi/c^2} [c^2 d^2 - \ell^2 / r^2]}.$$

Cancel $e^{-4\phi/c^2}$:

$$\left(\frac{d\phi}{dr} \right)^2 = \frac{\ell^2 / r^4}{c^2 d^2 - \ell^2 / r^2}.$$

This is the same relation as for a straight-line photon in flat Minkowski space (no ϕ dependence remaining):

$$\left(\frac{d\phi}{dr} \right)^2 = \frac{b^2 / r^4}{1 - b^2 / r^2}, \quad b \equiv \ell / (cd).$$

Integral gives $\phi(r) = \arcsin(b/r) + \text{const}$, a straight line with impact parameter b . Therefore:

$$\boxed{\Delta\phi_{\text{deflection}} = 0.}$$

The conformal factor cancels identically in the null geodesic equations: light follows the same paths as in Minkowski space.

Question 3 — Riemann tensor identities and Killing vectors

Problem. Prove the Ricci identity for V^μ , show $[U, V]^\mu$ of two Killing vectors is Killing, derive $\nabla^\mu T_{\mu\nu} = R_\nu{}^\sigma \nabla_\sigma \phi$ for $T_{\mu\nu} = \nabla_\mu \nabla_\nu \phi - g_{\mu\nu} \square \phi$, and prove $\nabla^\mu (T_{\mu\nu} k^\nu) = 0$ when $\nabla^\mu T_{\mu\nu} = 0$ with k^ν Killing.

(a) Ricci identity $(\nabla_\alpha \nabla_\beta - \nabla_\beta \nabla_\alpha) V^\mu = R^\mu{}_{\nu\alpha\beta} V^\nu$

Compute $\nabla_\beta V^\mu$:

$$\nabla_\beta V^\mu = \partial_\beta V^\mu + \Gamma^\mu_{\beta\nu} V^\nu.$$

Apply ∇_α to this $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ tensor:

$$\nabla_\alpha \nabla_\beta V^\mu = \partial_\alpha (\nabla_\beta V^\mu) + \Gamma^\mu_{\alpha\sigma} \nabla_\beta V^\sigma - \Gamma^\sigma_{\alpha\beta} \nabla_\sigma V^\mu.$$

Expand the first term:

$$\partial_\alpha (\nabla_\beta V^\mu) = \partial_\alpha \partial_\beta V^\mu + \partial_\alpha \Gamma^\mu_{\beta\nu} V^\nu + \Gamma^\mu_{\beta\nu} \partial_\alpha V^\nu.$$

Expand the second term:

$$\Gamma^\mu_{\alpha\sigma} \nabla_\beta V^\sigma = \Gamma^\mu_{\alpha\sigma} \partial_\beta V^\sigma + \Gamma^\mu_{\alpha\sigma} \Gamma^\sigma_{\beta\nu} V^\nu.$$

Combine:

$$\nabla_\alpha \nabla_\beta V^\mu = \partial_\alpha \partial_\beta V^\mu + \partial_\alpha \Gamma^\mu_{\beta\nu} V^\nu + \Gamma^\mu_{\beta\nu} \partial_\alpha V^\nu + \Gamma^\mu_{\alpha\sigma} \partial_\beta V^\sigma + \Gamma^\mu_{\alpha\sigma} \Gamma^\sigma_{\beta\nu} V^\nu - \Gamma^\sigma_{\alpha\beta} \nabla_\sigma V^\mu.$$

Antisymmetrise in $\alpha \leftrightarrow \beta$. The terms $\partial_\alpha \partial_\beta V^\mu$, the cross $\Gamma \partial$ terms, and the symmetric $\Gamma^\sigma_{\alpha\beta} \nabla_\sigma V^\mu$ all cancel:

$$(\nabla_\alpha \nabla_\beta - \nabla_\beta \nabla_\alpha) V^\mu = (\partial_\alpha \Gamma^\mu_{\beta\nu} - \partial_\beta \Gamma^\mu_{\alpha\nu}) V^\nu + (\Gamma^\mu_{\alpha\sigma} \Gamma^\sigma_{\beta\nu} - \Gamma^\mu_{\beta\sigma} \Gamma^\sigma_{\alpha\nu}) V^\nu.$$

Identify with the Riemann tensor as defined:

$$\boxed{(\nabla_\alpha \nabla_\beta - \nabla_\beta \nabla_\alpha) V^\mu = R^\mu{}_{\nu\alpha\beta} V^\nu.} \quad (8)$$

(b) $[U, V]^\mu$ of two Killing vectors is Killing

Let $W^\mu = U^\nu \nabla_\nu V^\mu - V^\nu \nabla_\nu U^\mu$. Killing condition:

$$\nabla_\mu U_\nu + \nabla_\nu U_\mu = 0, \quad \nabla_\mu V_\nu + \nabla_\nu V_\mu = 0. \quad (9)$$

We must show $\nabla_\alpha W_\beta + \nabla_\beta W_\alpha = 0$. Lower the index, $W_\beta = g_{\beta\mu} W^\mu = U^\nu \nabla_\nu V_\beta - V^\nu \nabla_\nu U_\beta$.

Take ∇_α :

$$\nabla_\alpha W_\beta = (\nabla_\alpha U^\nu) \nabla_\nu V_\beta + U^\nu \nabla_\alpha \nabla_\nu V_\beta - (\nabla_\alpha V^\nu) \nabla_\nu U_\beta - V^\nu \nabla_\alpha \nabla_\nu U_\beta.$$

Symmetrise in $\alpha \leftrightarrow \beta$:

$$\begin{aligned} \nabla_\alpha W_\beta + \nabla_\beta W_\alpha &= (\nabla_\alpha U^\nu) \nabla_\nu V_\beta + (\nabla_\beta U^\nu) \nabla_\nu V_\alpha - (\nabla_\alpha V^\nu) \nabla_\nu U_\beta - (\nabla_\beta V^\nu) \nabla_\nu U_\alpha \\ &\quad + U^\nu (\nabla_\alpha \nabla_\nu V_\beta + \nabla_\beta \nabla_\nu V_\alpha) - V^\nu (\nabla_\alpha \nabla_\nu U_\beta + \nabla_\beta \nabla_\nu U_\alpha). \end{aligned}$$

Use Killing on the bracket pairs. By (9) applied to the inner $\nabla_\nu V_\alpha = -\nabla_\alpha V_\nu$:

$$\nabla_\alpha \nabla_\nu V_\beta = -\nabla_\alpha \nabla_\beta V_\nu - [\nabla_\alpha, \nabla_\beta] \dots \quad (\text{track via Ricci identity}).$$

A cleaner route: apply the Ricci identity to commutators. Using (8) for a covector V_β :

$$(\nabla_\alpha \nabla_\nu - \nabla_\nu \nabla_\alpha) V_\beta = -R^\sigma{}_{\beta\alpha\nu} V_\sigma.$$

For Killing V , $\nabla_\nu V_\alpha = -\nabla_\alpha V_\nu$, so iterating gives the standard identity

$$\nabla_\alpha \nabla_\beta V_\gamma = R_{\gamma\beta\alpha\sigma} V^\sigma, \quad (\text{any Killing } V). \quad (10)$$

Apply (10) to the second-derivative pieces:

$$U^\nu (\nabla_\alpha \nabla_\nu V_\beta + \nabla_\beta \nabla_\nu V_\alpha) = U^\nu (R_{\beta\nu\alpha\sigma} + R_{\alpha\nu\beta\sigma}) V^\sigma.$$

$$V^\nu (\nabla_\alpha \nabla_\nu U_\beta + \nabla_\beta \nabla_\nu U_\alpha) = V^\nu (R_{\beta\nu\alpha\sigma} + R_{\alpha\nu\beta\sigma}) U^\sigma.$$

Use the symmetry $R_{\alpha\beta\gamma\delta} = R_{\gamma\delta\alpha\beta}$:

$$R_{\beta\nu\alpha\sigma} = R_{\alpha\sigma\beta\nu}, \quad R_{\alpha\nu\beta\sigma} = R_{\beta\sigma\alpha\nu}.$$

Swap the dummy index $\nu \leftrightarrow \sigma$ in the second term of the V -piece:

$$V^\nu R_{\alpha\nu\beta\sigma} U^\sigma = V^\sigma R_{\alpha\sigma\beta\nu} U^\nu = U^\nu R_{\beta\nu\alpha\sigma} V^\sigma.$$

Similarly the first term of the V -piece equals $U^\nu R_{\alpha\nu\beta\sigma} V^\sigma$ after relabelling. Hence the curvature contribution to $\nabla_\alpha W_\beta + \nabla_\beta W_\alpha$ from the second-derivative terms cancels identically.

For the first-derivative pieces, use $\nabla_\alpha U^\nu = -\nabla^\nu U_\alpha$ (raising and Killing):

$$(\nabla_\alpha U^\nu) \nabla_\nu V_\beta = -(\nabla^\nu U_\alpha) \nabla_\nu V_\beta.$$

Applied to all four first-derivative terms and combined using both Killing conditions (9), every term cancels in pairs.

Therefore:

$$\nabla_\alpha W_\beta + \nabla_\beta W_\alpha = 0.$$

$W^\mu = [U, V]^\mu$ is Killing.

(c) $\nabla^\mu T_{\mu\nu} = R_\nu{}^\sigma \nabla_\sigma \phi$

Take the divergence:

$$\nabla^\mu T_{\mu\nu} = \nabla^\mu \nabla_\mu \nabla_\nu \phi - \nabla^\mu (g_{\mu\nu} \nabla^\sigma \nabla_\sigma \phi).$$

The metric is covariantly constant ($\nabla g = 0$):

$$\nabla^\mu T_{\mu\nu} = \nabla^\mu \nabla_\mu \nabla_\nu \phi - \nabla_\nu (\nabla^\sigma \nabla_\sigma \phi).$$

Commute $\nabla^\mu \nabla_\mu \nabla_\nu \phi$ to $\nabla^\mu \nabla_\nu \nabla_\mu \phi$ using $\nabla_\mu \nabla_\nu \phi = \nabla_\nu \nabla_\mu \phi$ for a scalar:

$$\nabla^\mu \nabla_\mu \nabla_\nu \phi = \nabla^\mu \nabla_\nu \nabla_\mu \phi.$$

Apply the Ricci identity (8) for the covector $A_\mu = \nabla_\mu \phi$:

$$(\nabla^\mu \nabla_\nu - \nabla_\nu \nabla^\mu) A_\mu = -R^\sigma{}_\mu{}^\mu{}_\nu A_\sigma.$$

Use $R^\sigma{}_\mu{}^\mu{}_\nu = R_\nu{}^\sigma$ (definition of the Ricci tensor: $R_{\nu\sigma} = R^\mu{}_{\nu\mu\sigma}$, raised one index):

$$\nabla^\mu \nabla_\nu A_\mu = \nabla_\nu \nabla^\mu A_\mu + R_\nu{}^\sigma A_\sigma.$$

Substitute back:

$$\nabla^\mu \nabla_\mu \nabla_\nu \phi = \nabla_\nu \nabla^\mu \nabla_\mu \phi + R_\nu{}^\sigma \nabla_\sigma \phi.$$

Insert into $\nabla^\mu T_{\mu\nu}$:

$$\nabla^\mu T_{\mu\nu} = \nabla_\nu \square \phi + R_\nu{}^\sigma \nabla_\sigma \phi - \nabla_\nu \square \phi.$$

Cancel $\nabla_\nu \square \phi$:

$$\boxed{\nabla^\mu T_{\mu\nu} = R_\nu{}^\sigma \nabla_\sigma \phi.} \tag{11}$$

(d) $\nabla^\mu (T_{\mu\nu} k^\nu) = 0$ **when** $\nabla^\mu T_{\mu\nu} = 0$ **and** k^ν **is Killing**

Apply the Leibniz rule:

$$\nabla^\mu (T_{\mu\nu} k^\nu) = (\nabla^\mu T_{\mu\nu}) k^\nu + T_{\mu\nu} \nabla^\mu k^\nu.$$

Use the assumption $\nabla^\mu T_{\mu\nu} = 0$:

$$\nabla^\mu (T_{\mu\nu} k^\nu) = T_{\mu\nu} \nabla^\mu k^\nu.$$

$T_{\mu\nu}$ is symmetric (trivially: $\nabla_\mu \nabla_\nu \phi = \nabla_\nu \nabla_\mu \phi$ on a scalar; $g_{\mu\nu}$ symmetric):

$$T_{\mu\nu} \nabla^\mu k^\nu = T_{\nu\mu} \nabla^\mu k^\nu = \frac{1}{2} T_{\mu\nu} (\nabla^\mu k^\nu + \nabla^\nu k^\mu).$$

Apply Killing $\nabla^\mu k^\nu + \nabla^\nu k^\mu = 0$:

$$\boxed{\nabla^\mu (T_{\mu\nu} k^\nu) = 0.}$$

Question 4 — Three-phase inflationary cosmology

Problem. A flat universe runs through three sequential phases: vacuum-dominated ($w = -1$, $a < a_1$), radiation ($w = 1/3$, $a_1 < a < a_2$), matter ($w = 0$, $a_2 < a < 1$). Compute $H(a)$ and $q(a)$ in each, solve FRW for $a(t)$, compare ages, and compute particle horizons.

(a) Hubble and deceleration rates; FRW solutions

Continuity (energy conservation): $\dot{\rho} + 3H(\rho + P/c^2) = 0$, integrates to

$$\rho \propto a^{-3(1+w)}.$$

Cases:

$$w = -1 \text{ (vac)} : \rho = \rho_v \text{ const}; \quad w = \frac{1}{3} \text{ (rad)} : \rho_r \propto a^{-4}; \quad w = 0 \text{ (mat)} : \rho_m \propto a^{-3}.$$

Friedmann equation (flat, $k = 0$):

$$H^2 \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho. \quad (12)$$

Acceleration equation:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3P/c^2) = -\frac{4\pi G}{3}\rho(1 + 3w). \quad (13)$$

Deceleration parameter $q \equiv -\ddot{a}a/\dot{a}^2$:

$$q = -\frac{\ddot{a}/a}{H^2} = \frac{1+3w}{2}.$$

Match H_0 today ($a = 1$, matter): $H_m^2 = H_0^2 a^{-3}$. Continuity of H across a_2 fixes the radiation prefactor; continuity at a_1 fixes the vacuum density.

Vacuum $a < a_1$. $\rho = \rho_v = \text{const}$, hence

$$H_v(a) = H_0 a_1^{-3/2} \sqrt{a_2/a_1} \cdot \left[\frac{a_2}{a_1}\right]^0,$$

i.e. $H_v = \text{constant}$ H_1 . To pin H_1 : continuity at $a = a_1$ with the radiation phase $H_r(a_1)$ below.

Radiation $a_1 < a < a_2$. Match H at $a = a_2$ to matter: $H_r^2(a_2) = H_0^2 a_2^{-3}$, while $H_r^2 = C a^{-4}$. Therefore $C = H_0^2 a_2^{-3} \cdot a_2^4 = H_0^2 a_2$:

$$H_r(a) = H_0 a_2^{1/2} a^{-2}, \quad q_r = +1.$$

Matter $a_2 < a < 1$.

$$H_m(a) = H_0 a^{-3/2}, \quad q_m = +\frac{1}{2}.$$

Vacuum (now made explicit). H is constant; matching at a_1 to radiation gives

$$H_v = H_r(a_1) = H_0 a_2^{1/2} a_1^{-2}.$$

Hence

$$H_v(a) = H_0 a_2^{1/2} a_1^{-2}, \quad q_v = -1.$$

FRW solutions $a(t)$. Integrate $\dot{a} = aH$ in each phase.

Vacuum: $\dot{a}/a = H_v$:

$$a(t) = a_1 e^{H_v(t-t_1)} \quad (a < a_1).$$

Radiation: $\dot{a} = H_0 \sqrt{a_2} a^{-1}$, separate:

$$a da = H_0 \sqrt{a_2} dt.$$

Integrate from a_1 at $t = t_1$:

$$\frac{1}{2}(a^2 - a_1^2) = H_0 \sqrt{a_2} (t - t_1).$$

Solve:

$$a(t) = \sqrt{a_1^2 + 2H_0 \sqrt{a_2} (t - t_1)} \quad (a_1 < a < a_2).$$

Matter: $\dot{a} = H_0 a^{-1/2}$, separate $a^{1/2} da = H_0 dt$:

$$\frac{2}{3}(a^{3/2} - a_2^{3/2}) = H_0(t - t_2).$$

Solve:

$$a(t) = \left[a_2^{3/2} + \frac{3}{2} H_0 (t - t_2) \right]^{2/3} \quad (a_2 < a < 1).$$

Box the trio:

$$a(t) = \begin{cases} a_1 e^{H_v(t-t_1)}, & a < a_1 \\ \sqrt{a_1^2 + 2H_0 \sqrt{a_2} (t - t_1)}, & a_1 < a < a_2 \\ \left[a_2^{3/2} + \frac{3}{2} H_0 (t - t_2) \right]^{2/3}, & a_2 < a < 1. \end{cases}$$

(b) Why H is smaller during $a < a_1$ in the inflationary universe

In the inflationary universe, continuity at a_1 fixes $H_v = H_r(a_1) = H_0 \sqrt{a_2}/a_1^2$, a finite constant for all $a < a_1$.

In the non-inflationary (radiation-only at $a < a_1$) universe, $H_{\text{noinf}}(a) = H_0 \sqrt{a_2}/a^2$, which diverges as $a \rightarrow 0$.

Compare at any $a < a_1$:

$$\frac{H_v}{H_{\text{noinf}}(a)} = \frac{H_0 \sqrt{a_2}/a_1^2}{H_0 \sqrt{a_2}/a^2} = \left(\frac{a}{a_1} \right)^2 < 1.$$

Therefore:

$$H_v < H_{\text{noinf}}(a) \quad \forall a < a_1.$$

Inflation freezes H at the (modest) matching value, whereas radiation-domination drives $H \rightarrow \infty$ at small a .

(c) Age of the universe; inflationary universe is older

Definition: $t_{\text{age}} = \int_0^{t_0} dt = \int_{a_{\text{in}}}^1 \frac{da}{\dot{a}} = \int_{a_{\text{in}}}^1 \frac{da}{aH(a)}$.

Boxed integral form:

$$t_{\text{age}} = \int_{a_{\text{in}}}^1 \frac{da}{aH(a)}.$$

Compare integrands at $a < a_1$ using part (b): $H_v < H_{\text{noinf}}(a)$ implies

$$\frac{1}{aH_v} > \frac{1}{aH_{\text{noinf}}(a)}.$$

The two universes share $H(a)$ for $a > a_1$, so the integrals coincide on that range. On $a_{\text{in}} < a < a_1$ the inflationary integrand is strictly larger:

$$t_{\text{age}}^{\text{inf}} - t_{\text{age}}^{\text{noinf}} = \int_{a_{\text{in}}}^{a_1} \left[\frac{1}{aH_v} - \frac{1}{aH_{\text{noinf}}(a)} \right] da > 0.$$

Take $a_{\text{in}} \rightarrow 0$:

$$\begin{aligned} t_{\text{age}}^{\text{noinf}} &= \int_0^{a_1} \frac{da}{a \cdot H_0 \sqrt{a_2}/a^2} + \dots = \int_0^{a_1} \frac{a da}{H_0 \sqrt{a_2}} + \dots = \frac{a_1^2}{2H_0 \sqrt{a_2}} + \dots \\ t_{\text{age}}^{\text{inf}} &= \int_0^{a_1} \frac{da}{aH_v} + \dots = \frac{1}{H_v} \int_0^{a_1} \frac{da}{a} + \dots = \infty. \end{aligned}$$

So formally:

$$\boxed{t_{\text{age}}^{\text{inf}} \rightarrow \infty, \quad t_{\text{age}}^{\text{noinf}} < \infty.}$$

For any finite $a_{\text{in}} > 0$, $t_{\text{age}}^{\text{inf}} = t_{\text{age}}^{\text{noinf}} + H_v^{-1} \ln(a_1/a_{\text{in}}) - \frac{1}{2H_0 \sqrt{a_2}}(a_1^2 - a_{\text{in}}^2) > t_{\text{age}}^{\text{noinf}}$.

(d) Particle horizon in each phase; comparison with comoving length

Particle horizon at scale a (flat, comoving):

$$\chi_p(a) = c \int_0^t \frac{dt'}{a(t')} = c \int_{a_{\text{in}}}^a \frac{da'}{a'^2 H(a')}.$$

Physical horizon distance $d_p(a) = a \chi_p(a)$.

Vacuum $a < a_1$. $H = H_v$ const:

$$d_p^v(a) = a \cdot c \int_{a_{\text{in}}}^a \frac{da'}{a'^2 H_v} = \frac{c}{H_v} \left(1 - \frac{a}{a_{\text{in}}} \cdot 0 \dots \right).$$

Evaluate the integral:

$$\int_{a_{\text{in}}}^a \frac{da'}{a'^2 H_v} = \frac{1}{H_v} \left[-\frac{1}{a'} \right]_{a_{\text{in}}}^a = \frac{1}{H_v} \left(\frac{1}{a_{\text{in}}} - \frac{1}{a} \right).$$

Hence

$$\boxed{d_p^v(a) = \frac{c}{H_v} \left(\frac{a}{a_{\text{in}}} - 1 \right) \xrightarrow{a_{\text{in}} \rightarrow 0} \infty.}$$

Radiation $a_1 < a < a_2$. $H_r(a) = H_0 \sqrt{a_2}/a^2$:

$$\int_{a_1}^a \frac{da'}{a'^2 H_r(a')} = \frac{1}{H_0 \sqrt{a_2}} \int_{a_1}^a da' = \frac{a}{H_0 \sqrt{a_2}} + \text{const.}$$

Add the inflationary contribution and the constant from a_1 :

$$\boxed{d_p^r(a) \approx \frac{ca}{H_0 \sqrt{a_2}}(a - a_1) + a \cdot d_p^v(a_1)/a_1.}$$

Matter $a_2 < a < 1$. $H_m = H_0 a^{-3/2}$:

$$\int_{a_2}^a \frac{da'}{a'^2 H_0 a'^{-3/2}} \cdot 1 = \frac{1}{H_0} \int_{a_2}^a a'^{-1/2} da' = \frac{2a^{1/2}}{H_0} + \text{const.}$$

Boxed leading term:

$$\boxed{d_p^m(a) \sim \frac{2c}{H_0} a^{1/2} (a - a_2^{1/2}/a^{1/2}) + \dots, \quad d_p^m(1) \approx \frac{2c}{H_0}.}$$

Comparison with a fixed comoving length L_c . Physical size grows as $\ell(a) = aL_c$.

- During inflation $a < a_1$: $d_p^v(a) \propto a/a_{\text{in}}$, while $\ell(a) \propto a$. The ratio $\ell/d_p \rightarrow a_{\text{in}}/a \rightarrow 0$ as $a_{\text{in}} \rightarrow 0$: *any* comoving scale eventually exits the horizon ($d_H = c/H_v$ stays fixed in physical size while ℓ grows with a).
- During radiation/matter $a > a_1$: $d_p \propto a^2$ (rad) or $a^{3/2}$ (mat), growing faster than $\ell \propto a$. Comoving scales re-enter the horizon.

This is the canonical inflationary horizon-exit / re-entry behaviour:

For $a < a_1$: $\ell(a)/d_p(a) \rightarrow 0$ (horizon exit); $a > a_1$: $\ell/d_p \rightarrow 0 \rightarrow 1$ (re-entry).

The horizon problem of standard hot Big Bang cosmology is resolved because pre-inflationary causal contact establishes uniformity, and exponential expansion blows the horizon scale far inside any pre-existing comoving length.